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DOWNSTREAM PROCESSING OF EMBEDDING INFORMATION ITEMS

Abstract

A method for downstream processing of embedding information items, the method includes (i) receiving multiple evaluated element embedding information items that represent multiple evaluated elements within an environment of a vehicle; (ii) identifying that the multiple evaluated element embedding information items are classified into an insufficient confidence level; and (iii) for each one of the multiple evaluated embedding information items identified as an being classified into the insufficient confidence level, automatically routing evaluated element information to a corresponding embedding information item-based classification unit that is trained to classify elements represented by the evaluated element embedding information item associated with the corresponding population of embedding information items.

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Background/Summary

BACKGROUND

[0001] Assisted and autonomous driving systems are known in the art. In such systems, computer implemented systems control (at least to some extent) some, or all, of a vehicle's driving functions, e.g., speed, telemetry, braking, etc. The vehicle is typically equipped with one or more sensors to provide the system with current information regarding the driving environment. The current information for the driving environment is typically used by the driving system to determine how to drive on roadways.

[0002] One of the major tasks related to driving is classifying.

[0003] There is a growing need to provide efficient classification systems and methods.

SUMMARY

[0004] A method, system and non-transitory computer readable medium as illustrated in the application.

Description

A BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The embodiments of the disclosure will be understood and appreciated more fully from the following detailed description, taken in conjunction with the drawings in which:

[0006] FIG. 1A illustrates an example of a system;

[0007] FIG. 1B illustrates an example of a system;

[0008] FIG. 1C illustrates an example of a system;

[0009] FIG. 2 illustrates an example of clusters, scores and embeddings;

[0010] FIG. 3 is an example of a method;

[0011] FIG. 4 is an example of a method;

[0012] FIG. 5 illustrates an example of a system;

[0013] FIG. 6 is an example of a method;

[0014] FIG. 7 is an example of an image and various information elements;

[0015] FIG. 8 is an example of an image, keypoints and a cropped image;

[0016] FIG. 9 is an example of an image, a cropped image and various information elements;

[0017] FIG. 10 is an example of a method;

[0018] FIG. 11 is an example of images and a cropped images;

[0019] FIG. 12 is an example of an image, cropped images, a first neural network, various units and various information elements; and

[0020] FIG. 13 is an example of a method.

DETAILED DESCRIPTION

[0021] There is provided a method, a system and a computer readable medium that are adaptable and are configured to adjust to dynamically received information and solve classification errors by using multiple embedding-based classification units-some of which are tailored to classify elements that were not classified by other embedding-based classification units.

[0022] An embedding information item may be an embedding or a signature of an embedding.

[0023] For simplicity of explanation, some of the examples refer to embeddings. Any reference to an embedding should be applied, mutatis mutandis to any other embedding information item-such as but not limited to a signature of the embedding.

[0024] According to an embodiment, there is provided a method, a system and a non-transitory computer readable medium for downstream processing of embeddings that are identified to be classified into an insufficient confidence level.

[0025] The embedding represents an element selected of an object or a road scenario. The element was captured by a sensed information unit. The embedding may be generated based on a cropped sensed information unit. An embedding that is defined as being classified into an insufficient confidence level indicates that the element is not properly classified by the embedding-based classification unit.

[0026] According to an embodiment, the identification of an embedding as being classified into an insufficient confidence level triggers downstream processing of the embedding-by another embedding-based classification unit-for classifying the element represented by the embedding. The other embedding-based classification unit better fits (for example better trained or be tailored for) to classify the element.

[0027] According to an embodiment, there are multiple other embedding-based classification units that are trained (or otherwise configured) to successfully classify an element that was classified into an insufficient confidence level.

[0028] According to an embodiment the embedding-based classification units are arranged in a hierarchical layer. According to an embodiment-an embedding-based classification unit of a higher level of the hierarchy is trained across a broader population of embeddings than an embedding-based classification unit of a lower level of the hierarchy.

[0029] According to an embodiment, there are provided routing rules that route the sensed information unit associated with the embedding being classified into an insufficient confidence level to a corresponding other embedding-based classification unit that is trained (or otherwise configured) to successfully classify the object.

[0030] The different figures illustrates examples of units and/or software and/or information items and/or steps and/or components. These examples are provided for brevity of explanation. At least one of the units and/or software and/or information items and/or steps and/or components is optional or mandatory.

[0031] FIG. 1A illustrates an example of a vehicle **100**, a network **123** and remote computerized systems **134**.

[0032] The vehicle **100** includes (a) sensing system **110**, a communication system **130**, one or more memory and/or storage units **120**, and additional units that include control unit **125**, advanced driver assistance system (ADAS) control unit **123**, autonomous driving control unit **122**, processing system **124** including processor **126**. Network **123** is in communication with the vehicle and with the remote computerized systems **134** such as servers, cloud computers, and the like.

[0033] Communication system **130**, one or more memory and/or storage units **120**, and processing system **134** may form a computerized system. The computerized system may include one or more other systems and/or units such as sensing system **110**

[0034] The communication system **130** is configured to enable communication between the one or more memory and/or storage units **120** and/or the sensing system **110** and/or any one of the additional units and/or the network **132** (that is in communication with the remote computerized systems).

[0035] The control unit **125** is configured to control various operations related to the vehicle-such as but not limited to various steps of method **200** and/or of method **201**.

[0036] The one or more memory and/or storage units **120** are illustrated as storing an operating system **194**, software **193** (especially software required to execute method **200** and/or of method **201**), information **191** and metadata **192** (especially information and metadata required to execute method **200** and/or of method **201**). The information may include environmental information. The metadata may include any metric or an outcome of processed information-especially related to the execution of method **200** and/or of method **201**.

[0037] FIG. 1B and FIG. 1C differ from FIG. 1A by including additional units such as ADAS control unit **123**, autonomous driving control unit **123**, and vehicle computer **123**.

[0038] FIG. 1B and FIG. 1C differ from FIG. 1A by including more examples of content stored in

the one or more memory and/or storage units **120**.

[0039] The sensing system **110** may include optics, a sensing element group, a readout circuit, and an image signal processor. Optics are followed by a sensing element group such as line of sensing elements or an array of sensing elements that form the sensing element group. The sensing element group is followed by a readout circuit that reads detection signals generated by the sensing element group. An image signal processor is configured to perform an initial processing of the detection signals—for example by improving the quality of the detection information, performing noise reduction, and the like. The sensing system **110** is configured to output one or more sensed information units (SIUs).

[0040] The communication system **130** is configured to enable communication between the one or more memory and/or storage units **120** and/or the sensing system **110** and/or any one of the additional units and/or the network **132** (that is in communication with the remote computerized systems).

[0041] The controller **125** is configured to control the operation of the sensing system **110**, and/or the one or more memory and/or storage units **120** and/or the one or more additional units (except the controller).

[0042] The ADAS control unit **123** is configured to control ADAS operations.

[0043] The autonomous driving control unit **122** is configured to control autonomous driving of the autonomous vehicle.

[0044] The vehicle computer **121** is configured to control the operation of the vehicle—especially controlling the engine, the transmission, and any other vehicle system or component.

[0045] The processing system **124** may include processor **146** and one or more other processors and is configured to execute any method illustrated in the specification.

[0046] The one or more memory and/or storage units **120** are configured to store firmware and/or software, one or more operating systems, data and metadata required to the execution of any of the methods mentioned in this application.

[0047] FIGS. **1B** and/or FIG. **1C** illustrates the one or more memory and/or storage units **120** as storing at least some of: [0048] a. Multiple evaluated elements embeddings **161**. [0049] b.

Evaluated element information **162** such as sensed information units (for example—cropped or not cropped sensed information units). [0050] c. Outliers **163**. [0051] d. Reference embeddings **164**. [0052] e. Reference embeddings clusters **165**. [0053] f. Known element classifications **166**. [0054] g. Response software **167** that may include routing software **167-1**, a re-evaluation software **167-2** for re-evaluating the plurality of reference embeddings. [0055] h. Embedding-based classification software **171**. The embedding-based classification software may include embedding generation software **171-1** and a matching software **171-2** configured to match an evaluated element embedding to the reference embeddings. [0056] i. Additional embedding-based classification software **172**. [0057] j. Routing rules **168**. [0058] k. Signature generation software **174** configured to generate a signature. The signature may be a signature of an embedding. The signature may be generated based on the SIU but may be generated based on other information. Examples of a signature and signature generators that are examples of detectors are illustrated in U.S. patent application Ser. No. **17/309,064** publication serial number **2022/0041184** which is incorporated herein by reference. [0059] l. Operating system **169-1**. [0060] m. Additional software **169-2** that may be used to perform any other functionality of the vehicle and/or of any of the other units illustrated in FIG. **1**.

[0061] The vehicle computer **121** may be in communication with an engine control module, a transmission control module, a powertrain control module, and the like

[0062] The memory and/or storage units **120** was shown as storing software. Any reference to software should be applied mutatis mutandis to code and/or firmware and/or instructions and/or commands, and the like.

[0063] Processor **126** includes a plurality of processing units **126(1)-126(J)**, J is an integer that

exceeds one. Any reference to one unit or item should be applied mutatis mutandis to multiple units or items. For example-any reference to processor should be applied mutatis mutandis to multiple processors, any reference to communication system **130** should be applied mutatis mutandis to multiple communication systems.

[0064] According to an embodiment, the one or more memory and/or storage units **120** includes one or more memory unit, each memory unit may include one or more memory banks.

[0065] According to an embodiment, the one or more memory and/or storage units **120** includes a volatile memory and/or a non-volatile memory. The one or more memory and/or storage units **120** may be a random-access memory (RAM) and/or a read only memory (ROM).

[0066] According to an embodiment, the non-volatile memory unit is a mass storage device, which can provide non-volatile storage of computer code, computer readable instructions, data structures, program modules, and other data for the processor or any other unit of vehicle. For example, and not meant to be limiting, a mass storage device can be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.

[0067] Any content may be stored in any part or any type of the memory and/or storage units.

[0068] According to an embodiment, the at least one memory unit stores at least one database—such as any database known in the art-such as DB2®, Microsoft® Access, Microsoft® SQL Server, Oracle®, mySQL, PostgreSQL, and the like.

[0069] Various units and/or components are in communication with each other using any communication elements and/or protocols. An example of a communication system is denoted **130**. Other communication elements may be provided.

[0070] FIG. **1** illustrates communication system **130** as being in communication with various processors and/or units and network **132**.

[0071] The communication system **130** may include a bus. The represents one or more of several possible types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, such architectures can comprise an Industry Standard

[0072] Architecture (ISA) bus, a Micro Channel Architecture (MCA) bus, an Enhanced ISA (EISA) bus, a Video Electronics Standards Association (VESA) local bus, an Accelerated Graphics Port (AGP) bus, and a Peripheral Component Interconnects (PCI), a PCI-Express bus, a Personal Computer Memory Card Industry Association (PCMCIA), Universal Serial Bus (USB) and the like. The bus, and all buses specified in this description can also be implemented over a wired or wireless network connection and each of the subsystems.

[0073] Network **132** that is located outside the vehicle and is used for communication between the vehicle and at least one remote computing system. By way of example, a remote computing system can be a personal computer, a laptop computer, portable computer, a server, a router, a network computer, a peer device or other common network node, and so on. Logical connections between the processor and either one of remote computing systems can be made via a local area network (LAN) and a general wide area network (WAN). Such network connections can be through a network adapter (may belong to communication system **130**) which can be implemented in both wired and wireless environments. Such networking environments are conventional and commonplace in offices, enterprise-wide computer networks, intranets, and a larger network such as the internet.

[0074] It should be noted that at least a part of the content illustrated as being stored in one or more memory/storage units **120** may be stored outside the vehicle. It should also be noted that the processor may evaluate signatures generated by a plurality of detectors.

[0075] According to an embodiment-an evaluated element embedding information item may be

classified with a confidence level. The confidence level may be binary (outlier or inlier) or non-binary—for example may be a percent; or score; or any level within a specified range; or any level out of two or more levels. The confidence level may be calculated in various manners—such as based on a distance between the evaluated element embedding information to a reference point—such as a centroid of a cluster—or any other reference point—such as a reference point located within (or without) the cluster.

[0076] Any reference to an embedding information item that is classified into an insufficient confidence level should be applied *mutatis mutandis* to an outlier.

[0077] According to an embodiment, what amount to insufficient confidence level is defined in one or more manners and/or by one or more entities (such as a user, a driver, a manufacturer, an insurance company, a vendor, and the like). For example—an inlier may be associated with an insufficient confidence level if he is located at least a defined distance away from the centroid and/or if is among a defined percent of the most distant inliers from the centroid, and the like. If, for example the level of confidence level ranges between 1 to 100 then what amount to insufficient may be lower than any number—for example lower than 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 98, 99 and the like. According to an embodiment, the processor is configured to: [0078] a. Receive multiple evaluated elements embeddings that represent multiple evaluated elements within an environment of a vehicle. An evaluated element embedding is generated based on evaluated element information such as a sensed information unit or a cropped sensed information unit. [0079] b. Evaluate whether the multiple evaluated elements embeddings are classified into an insufficient confidence level. According to an embodiment, the evaluation includes comparing each evaluated element embedding (of the multiple evaluated elements embeddings) to a plurality of reference embeddings. The plurality of reference embeddings represents a plurality of reference embeddings clusters. The plurality of reference embeddings clusters is associated with a plurality of known element classifications. It should be noted that the processing circuit may also receive evaluated element embeddings that are inliers—that represent evaluated elements of known classes. In this case, step **220** may include or may be followed by successfully classifying the evaluated elements associated with the evaluated embeddings. [0080] c. Responding to the identification.

[0081] According to an embodiment, the responding includes at least one of: [0082] a. Automatically routing evaluated element information associated with each evaluated element embedding of the multiple evaluated element embeddings, to a corresponding embedding-based classification unit that is trained to classify evaluated elements represented by the evaluated element embedding. This may include selecting each corresponding embedding-based classification unit out of a plurality of embedding-based classification units. [0083] b. Classify the evaluated element, by the corresponding embedding-based classification unit. According to an embodiment, the classifying triggers a determination of a driving related operation for a vehicle. The driving related operation is selected out of performing an autonomous driving operation, performing an advanced driver assistance system (ADAS) operation, and the like. According to an embodiment, the classifying triggers an execution of a driving related operation for a vehicle.

[0084] c. Triggering a generation of a routing rule for bypassing the classifying when receiving future evaluated element embeddings having a same value as the evaluated element embedding previously classified as being classified into an insufficient confidence level. [0085] d. Triggering a re-evaluation of the plurality of reference embeddings. [0086] e. Determine that there is no existing embedding-based classification unit configured to classify an evaluated element represented by an evaluated element embedding identified as being classified into an insufficient confidence level. This may trigger a generation of a new embedding-based classification unit.

[0087] FIG. 2 illustrates an example of a first embedding-based classification unit **180**, a routing unit **181**, outliers **191'** and other embedding-based classification units **182-1** till **182-K**.

[0088] Once an evaluated element embedding is found (by the first embedding-based classification

unit **180**) to be being classified into an insufficient confidence level, the routing unit **181** routes the evaluated element information **192** (associated with the evaluated element embedding **191**) to one of the other embedding-based classification units **182-1** till **182-K**.

[0089] The routing may be based on a similarity between the evaluated element embedding and embeddings that represent the other embedding-based classification units **182-1** till **182-K**.

[0090] FIG. **2** also illustrates reference embedding clusters **185-1** till **185-5**. Referring to FIG. **2**—cluster **185-1** includes centroid **185-1-0** and various inliers such as **185-1-1** and **185-1-2** (and may also include other inliers that are not shown).

[0091] According to an embodiment, inlier **185-1-1** is closer to the centroid **185-1-0** in relation to inlier **185-1-2** and may be associated with a higher confidence level. Scores **199** of FIG. **2** are example of the confidence levels of various inliers.

[0092] FIG. **3** illustrates an example of method **200** downstream processing of embedding information items.

[0093] According to an embodiment, method **200** includes step **210** of receiving, by a processing circuit, multiple evaluated elements embeddings that represent multiple evaluated elements within an environment of a vehicle. An evaluated element embedding is generated based on evaluated element information such as a sensed information unit or a cropped sensed information unit.

[0094] According to an embodiment, step **210** is followed by step **220** of identifying that the multiple evaluated elements embeddings classified into an insufficient confidence level. A population of embeddings includes one or more clusters of inliers and one or more outliers.

[0095] According to an embodiment, step **220** includes comparing each evaluated element embedding (of the multiple evaluated elements embeddings) to a plurality of reference embeddings. The plurality of reference embeddings represents a plurality of reference embeddings clusters. The plurality of reference embeddings clusters is associated with a plurality of known element classifications.

[0096] It should be noted that the processing circuit may also receive evaluated element embeddings that are inliers—that represent evaluated elements of known classes. In this case, step **220** may include or may be followed by successfully classifying the evaluated elements associated with the evaluated embeddings.

[0097] According to an embodiment, step **220** is followed by step **230** of responding to the identification.

[0098] According to an embodiment, step **230** includes step **231** of automatically routing evaluated element information associated with each evaluated element embedding of the multiple evaluated element embeddings, to a corresponding embedding-based classification unit that is trained to classify evaluated elements represented by the evaluated element embedding. These elements may belong to a population that includes the evaluated elements.

[0099] According to an embodiment, step **231** includes selecting each corresponding embedding-based classification unit out of a plurality of embedding-based classification units.

[0100] According to an embodiment, step **231** is followed by step **240** of classifying the evaluated element, by the corresponding embedding-based classification unit.

[0101] According to an embodiment, step **240** triggers a determination of a driving related operation for a vehicle. The driving related operation is selected out of performing an autonomous driving operation, performing an advanced driver assistance system (ADAS) operation, and the like.

[0102] According to an embodiment, step **240** triggers an execution of a driving related operation for a vehicle.

[0103] According to an embodiment, step **230** includes step **232** of triggering a generation of a routing rule for bypassing step **220** when receiving future evaluated element embeddings having a same value as the evaluated element embedding previously classified as being classified into an insufficient confidence level.

[0104] According to an embodiment, step **230** includes step **233** of triggering a re-evaluation of the plurality of reference embeddings.

[0105] According to an embodiment, step **230** includes step **234** of determining that there is no existing embedding-based classification unit configured to classify an evaluated element represented by an evaluated element embedding identified as being classified into an insufficient confidence level.

[0106] According to an embodiment, step **234** triggers a generation of a new embedding-based classification unit.

[0107] FIG. **4** illustrates an example of method **201** downstream processing of embedding signatures.

[0108] According to an embodiment, method **201** includes step **211** of receiving, by a processing circuit, multiple evaluated elements embeddings signatures that represent multiple evaluated elements within an environment of a vehicle. An evaluated element embedding signature is generated based on evaluated element information such as a sensed information unit or a cropped sensed information unit.

[0109] According to an embodiment, step **211** is followed by step **221** of identifying that the multiple evaluated elements embeddings signatures are classified into an insufficient confidence level.

[0110] According to an embodiment, step **221** includes comparing each evaluated element embedding signature (of the multiple evaluated elements embeddings signatures) to a plurality of reference embeddings signatures. The plurality of reference embeddings signatures represents a plurality of reference embeddings signatures clusters. The plurality of reference embeddings signatures clusters is associated with a plurality of known element classifications.

[0111] It should be noted that the processing circuit may also receive evaluated element embeddings signatures that are inliers-that represent evaluated elements of known classes. In this case, step **221** may include or may be followed by successfully classifying the evaluated elements associated with the evaluated embeddings signatures.

[0112] According to an embodiment, step **221** is followed by step **250** of responding to the identification.

[0113] According to an embodiment, step **250** includes step **251** of automatically routing evaluated element information associated with each evaluated element embedding signature of the multiple evaluated element embeddings signatures, to a corresponding embedding-based classification unit that is trained to classify evaluated elements represented by the evaluated element embedding.

[0114] According to an embodiment, step **250** includes selecting each corresponding embedding-based classification unit out of a plurality of embedding-based classification units.

[0115] According to an embodiment, step **250** is followed by step **260** of classifying the evaluated element, by the corresponding embedding-based classification unit.

[0116] According to an embodiment, step **250** triggers a determination of a driving related operation for a vehicle. The driving related operation is selected out of performing an autonomous driving operation, performing an advanced driver assistance system (ADAS) operation, and the like.

[0117] According to an embodiment, step **250** triggers an execution of a driving related operation for a vehicle.

[0118] According to an embodiment, step **251** includes step **252** of triggering a generation of a routing rule for bypassing step **221** when receiving future evaluated element embeddings signatures having a same value as the evaluated element embedding signature previously classified as being classified into an insufficient confidence level.

[0119] According to an embodiment, step **251** includes step **253** of triggering a re-evaluation of the plurality of reference embeddings signatures.

[0120] According to an embodiment, step **251** includes step **254** of determining that there is no

existing embedding-based classification unit configured to classify an evaluated element represented by an evaluated element embedding signature identified as being classified into an insufficient confidence level.

[0121] According to an embodiment, step **254** triggers a generation of a new embedding—based classification unit.

Examples of Embedding-Based Classification Units and Embedding-Based Classification Methods

[0122] FIG. **5** is a block diagram illustrating an exemplary operating environment for performing at least a portion of disclosed methods according to an embodiment of the present invention. This exemplary operating environment is only an example of an operating environment and is not intended to suggest any limitation as to the scope of use or functionality of operating environment architecture. Neither should the operating environment be interpreted as having any dependency or requirement relating to any one or combination of components illustrated in the exemplary operating environment.

[0123] Further, one skilled in the art will appreciate that the systems and methods disclosed herein can utilize a specialized computing device in the form of an object classification system computer **701** (which may be included in, for example object classification system **100**). The methods discussed above can be performed by the computer **701**. For example, the computer **701** can perform the duties and responsibilities discussed above.

[0124] The components of the object classification system computer **701** can comprise, but are not limited to, one or more processors or processing units **703**, a system memory **712**, and a system bus **713** that couples various system components including the processor **703** to the system memory **712**. In the case of multiple processing units **703**, the system can utilize parallel computing.

[0125] The system bus **713** represents one or more of several possible types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, such architectures can comprise an Industry Standard Architecture (ISA) bus, a Micro Channel Architecture (MCA) bus, an Enhanced ISA (EISA) bus, a Video Electronics Standards Association (VESA) local bus, an Accelerated Graphics Port (AGP) bus, and a Peripheral Component Interconnects (PCI), a PCI-Express bus, a Personal Computer Memory Card Industry Association (PCMCIA), Universal Serial Bus (USB) and the like. The bus **713**, and all buses specified in this description can also be implemented over a wired or wireless network connection and each of the subsystems, including the processor **703**, a mass storage device **704**, an operating system **705**, object classification system software **706**, object classification system data **707**, a network adapter **708**, system memory **712**, an Input/Output Interface **710**, a display adapter **709**, a display device **711**, and a human machine interface **702**, can be contained within one or more remote computing devices **714** a,b,c at physically separate locations, connected through buses of this form, in effect implementing a fully distributed system.

[0126] The object classification system computer **701** typically comprises a variety of computer readable media. Exemplary readable media can be any available media that is accessible by the object classification system computer **701** and comprises, for example and not meant to be limiting, both volatile and non-volatile media, removable and non-removable media. The system memory **712** comprises computer readable media in the form of volatile memory, such as random-access memory (RAM), and/or non-volatile memory, such as read only memory (ROM). The system memory **712** typically contains data such as object classification system data **707** and/or program modules such as operating system **705** and object classification system software **706** (i.e., modules and the like that perform the methods discussed above) that are immediately accessible to and/or are presently operated on by the processing unit **703**.

[0127] In another aspect, the object classification system computer **701** can also comprise other removable/non-removable, volatile/non-volatile computer storage media. By way of example, FIG. **7** illustrates a mass storage device **704**, which can provide non-volatile storage of computer code,

computer readable instructions, data structures, program modules, and other data for the object classification system computer **701**. For example, and not meant to be limiting, a mass storage device **704** can be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.

[0128] Optionally, any number of program modules can be stored on the mass storage device **704**, including by way of example, an operating system **705** and object classification system software **706**. Each of the operating system **705** and object classification system software **706** (or some combination thereof) can comprise elements of the programming and the object classification system software **706**. object classification system data **707** can also be stored on the mass storage device **704**. object classification system data **707** can be stored in any of one or more databases known in the art. Examples of such databases include DB2®, Microsoft® Access, Microsoft® SQL Server, Oracle®, MySQL, PostgreSQL, and the like. The databases can be centralized or distributed across multiple systems. In other aspects, the object classification system data **707** can be stored on the mass storage device **705** of other servers or devices (e.g., remote computing device **714 a,b,c**), in communication with the object classification system computer **701**.

[0129] In another aspect, the user can enter commands and information into the object classification system computer **701** via an input device (not shown). Examples of such input devices comprise, but are not limited to, a keyboard, pointing device (e.g., a “mouse”), a microphone, a joystick, a scanner, tactile input devices such as gloves, and other body coverings, and the like. These and other input devices can be connected to the processing unit **703** via a human machine interface **702** that is coupled to the system bus **713** but can be connected by other interface and bus structures, such as a parallel port, game port, an IEEE 1394 Port (also known as a Firewire port), a serial port, or a universal serial bus (USB).

[0130] In yet another aspect, a display device **711** can also be connected to the system bus **713** via an interface, such as a display adapter **709**. It is contemplated that the object classification system computer **701** can have more than one display adapter **709** and more than one display device **711**. For example, a display device can be a monitor, an LCD (Liquid Crystal Display), or a projector. In addition to the display device **711**, other output peripheral devices can comprise components such as speakers (not shown) and a printer (not shown) which can be connected to the computer **701** via Input/Output Interface **710**. Any step and/or result of the methods can be output in any form to an output device. Such output can be any form of visual representation, including, but not limited to, textual, graphical, animation, audio, tactile, and the like.

[0131] The object classification system computer **701** can operate in a networked environment using logical connections to one or more remote computing devices **714 a, b, c**. By way of example, a remote computing device can be a personal computer, a laptop computer, portable computer, a server, a router, a network computer, a peer device or other common network node, and so on. Logical connections between the object classification system computer **701** and a remote computing device **714 a, b, c** can be made via a local area network (LAN) and a general wide area network (WAN). Such network connections can be through a network adapter **708**. A network adapter **708** can be implemented in both wired and wireless environments. Such networking environments are conventional and commonplace in offices, enterprise-wide computer networks, intranets, and a network **715** such as the internet **715**.

[0132] For purposes of illustration, application programs and other executable program components such as the operating system **705** are illustrated herein as discrete blocks, although it is recognized that such programs and components reside at various times in different storage components of the object classification system computer **701** and are executed by the data processor(s) of the computer. An implementation of object classification system software **706** can be stored on or transmitted across some form of computer readable media. Any of the disclosed

methods can be performed by computer readable instructions embodied on computer readable media. Computer readable media can be any available media that can be accessed by a computer. By way of example and not meant to be limiting, computer readable media can comprise “computer storage media” and “communications media.” “Computer storage media” comprise volatile and non-volatile, removable and non-removable media implemented in any methods or technology for storage of information such as computer readable instructions, data structures, program modules, or other data. Exemplary computer storage media comprises, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information, and which can be accessed by a computer.

[0133] According to an embodiment, object classification system computer **701** is configured to execute any method illustrated in the application.

[0134] According to an embodiment the object classification system computer **701** is in communication with one or more sensors of one or more types that are associated with the vehicle.

[0135] According to an embodiment the object classification system computer **701** is in communication with other vehicle computes such as control computers that are configured to control one or more vehicle units such as an engine controlling computer, a powertrain controlling computer, and/or with an autonomous driving unit configured to control autonomous driving, an ADAS unit configured to control ADAS operations, a path unit configured to navigate the vehicle, and the like. Each unit includes a processing circuit and/or stores in a non-transitory computer readable medium software and/or firmware and/or code and/or instructions for fulfilling the role of the unit.

[0136] FIG. **6** illustrates an example of method **800**.

[0137] According to an embodiment method **800** is for object classification.

[0138] According to an embodiment, method **800** includes step **810** of receiving, by a processing circuit, a cropped sensed information unit that includes information indicative of an object located within an environment of a vehicle.

[0139] According to an embodiment, step **810** is followed by step **820** of generating, by the processing circuit, an object embedding information item representing the object.

[0140] An embedding information item may be an embedding or a representation (for example signature) of the embedding.

[0141] According to an embodiment, step **820** is followed by step **830** of comparing the object embedding information item to a plurality of reference embeddings information items that represent reference embedding information items clusters.

[0142] According to an embodiment, step **830** is followed by step **840** of identifying, based on the comparing step, a matching reference embedding information item that represents a matching reference embedding information items cluster.

[0143] According to an embodiment, step **840** is followed by step **850** of classifying the object as being associated with an object classification that is associated with the matching reference embedding information item cluster.

[0144] According to an embodiment, step **850** is followed by step **860** of responding to the classifying.

[0145] According to an embodiment, step **860** may include at least one of: [0146] a. Triggering a determination of a driving related operation to be executed by the vehicle. The driving related operation may be a fully autonomous operation such as maintaining a current progress of a vehicle, changing a progress of the vehicle, whereas a progress refers to at least one of direction of propagation, speed of propagation and acceleration of the vehicle. Alternatively-the driving related operation is an ADAS operation such as suggesting a driving scheme operation of performing a more limited driving related operation (limited in the sense of its duration). [0147] b. Executing a

driving related operation. [0148] c. Determining the navigation path of the vehicle. [0149] d. Populating a database with an outcome of the vehicle. [0150] e. Sending to one or more artificial intelligence agents information regarding the classification. [0151] f. Activating one or more artificial intelligence agents based on the classification. [0152] g. Further processing an initial (pre-cropping) sensed information unit. The cropped information unit was generated by cropping an initial (pre-cropping) information sensing unit.

[0153] According to an embodiment, the object embedding information item is an object embedding signature. The reference embeddings information items are reference embeddings signatures. The reference embedding information items clusters are reference embedding signatures clusters.

[0154] According to an embodiment, a signature of an embedding is of a higher dimension (also referred to a higher dimensionality or having more dimensions) than the embedding. The increase of the dimensionality may increase the robustness of the detection as higher dimensionality increases the distance between adjacent signatures—in comparison to the distance between corresponding adjacent embeddings. The higher dimension may include having at least 2, 5, 10, 20, 50, 80, 100 dimensions and even more.

[0155] A non-limiting example of generating a signature is illustrated in US patent application 2022/0041184 which is incorporated herein by reference.

[0156] According to an embodiment, step **820** includes step **821** of generating an object embedding and step **822** of generating the signature of the object embedding, wherein the object embedding has less dimensions than the object embedding signature.

[0157] FIG. 7 illustrates an example of an initial sensed information unit such as an image **901** including a bounding box **902** that is indicative of a location of object **903**, a cropped image **904** that mostly includes the pixels of the object **903**, an object embedding **910** of the cropped image, a signature **912** of the object embedding, reference embedding signatures clusters **920(1)-920(W)** that are represented by reference embedding signatures **921(1)-921(X)** (that may or may not be included in the reference embedding signature clusters), wherein there may be more than one reference embedding signature per reference embedding signature cluster, other reference embedding signatures **922(1)-922(Y)** included in the reference embedding signatures clusters (and differ from the reference embedding signatures **921(1)-921(X)**), outliers **925(1)-925(Z)** located outside the clusters, reference embedding signatures clusters metadata (**923(1)-923(W)**) that provide information such as object classification about the clusters, and a matching reference embedding signatures cluster **920(w)** that is represented by reference embedding signature **921(w)** and is associated with metadata **923(w)** indicative of an object classification.

[0158] According to an embodiment, the object embedding information item is an object embedding, the reference embeddings information items are reference embeddings, and the reference embedding information items clusters are reference embedding clusters.

[0159] According to an embodiment the cropped sensed information unit consists essentially of the information indicative of the object. The cropping increases the accuracy of the object embedding as irrelevant information is mostly removed from the cropped sensed information unit.

[0160] According to an embodiment, the cropped sensed information unit was generated based on an initial sensed information unit and a bounding box indicative of the object within the initial sensed information unit.

[0161] According to an embodiment, the cropped sensed information was generated based on an initial sensed information unit and a plurality of keypoints within the initial sensed information unit that are associated with the object. Keypoints may be found in any known manner. According to an embodiment, a keypoint is found in the manner illustrated in U.S. Pat. No. 11,037,015 which is incorporated herein by reference.

[0162] According to an embodiment, cropped sensed information was generated based on an initial sensed information unit and an initial sensed information unit region that includes a plurality of

keypoints within the initial sensed information unit that are associated with the object.

[0163] Assuming, that the transformation module (especially an embedding generator within the transformation module) is configured to process content of a given shape—for example it is configured to process a cropped sensed information unit that includes relevant information within a rectangular region of a sensed information unit—then the method includes defining a rectangular region to include the plurality of keypoints (the rectangular region may be defined to be as small as possible—or to include up a limited amount of information outside a smallest region that includes the keypoints)—and to process the content of the rectangular region. The processing may include aligning the rectangular region (which may be oriented to the horizon) before providing the rectangular box to the embedding generator.

[0164] FIG. 8 illustrates a sensed information unit such as image 950, including object 903, including keypoints 930 of the object, a bounding box 932 that surrounds the keypoints 930, and a cropped image 904. The bounding box 932 is oriented and the cropped image includes an aligned bounding box 933.

[0165] FIG. 9 illustrates an example of an initial sensed information unit such as an image 901 including a bounding box 902 that is indicative of a location of object 903, a cropped image 904 that mostly includes the pixels of the object 903, an object embedding 910 of the cropped image, reference embedding clusters 970(1)-970(W) that are represented by reference embedding 971(1)-971(X), wherein there may be more than one reference embedding per reference embedding cluster, other reference embedding 972(1)-972(Y) included in the reference embedding clusters (and differ from the reference embedding 971(1)-971(X)), outliers 975(1)-975(Z) located outside the clusters, reference embedding clusters metadata (973(1)-973(W)) that provide information such as object classification about the clusters, and a matching reference embedding cluster 970(w) that is represented by reference embedding 971(w).

[0166] FIG. 10 illustrates an example of method 801.

[0167] According to an embodiment method 80' is for object classification.

[0168] According to an embodiment, method 80' includes step 810 of receiving, by a

[0169] processing circuit, a cropped sensed information unit that includes information indicative of an object located within the environment of a vehicle.

[0170] According to an embodiment, method 810 also includes step 812 of receiving, by the processing circuit, an additional cropped sensed information unit that includes additional information indicative of the object located within the vehicle environment. While the cropped sensed information unit received in step 810 was sensed by a sensor of a first type, the additional cropped sensed information unit was sensed by a sensor of second type differs from the first type. The types of sensors may differ from each other by radiation frequency (for example radar versus visible light), resolution, point of view (for example aerial point of view or ground level point of view), active sensor versus passive sensor, and the like. Examples of sensors of different types include a visual light camera, an audio sensor, a sensor that may sense infrared, radar imagery, ultrasound, electro-optics, radiography, LIDAR (light detection and ranging), and the like.

[0171] According to an embodiment, step 810 and step 812 are followed by step 820 of generating, by the processing circuit, an object embedding information item representing the object.

[0172] According to an embodiment, step 820 is followed by step 830 of comparing the object embedding information item to a plurality of reference embeddings information items that represent reference embedding information items clusters.

[0173] According to an embodiment, step 830 is followed by step 840 of identifying, based on the comparing step, a matching reference embedding information item that represents a matching reference embedding information items cluster.

[0174] According to an embodiment, step 840 is followed by step 850 of classifying the object as being associated with an object classification that is associated with the matching reference embedding information item cluster.

[0175] According to an embodiment, step **850** is followed by step **860** of responding to the classifying.

[0176] FIG. **13** illustrates an example of an initial sensed information unit such as an image **901** including a bounding box **902** that is indicative of a location of object **903**, a cropped image **904** that mostly includes the pixels of the object **903**, an additional initial sensed information unit such as an additional image **901-1** including a bounding box **902-1** that is indicative of a location of object **903**, an additional cropped image **904-4** that mostly includes the pixels of the object **903**. The additional image is acquired by a sensor of a different type than the sensor that sensed image **901**.

[0177] FIG. **14** illustrates an example of a system **101** that includes a first neural network **1000-1** configured to receive a sensed information unit such as image **1010-1** and perform an object detection to provide an first output image **1010-2** with first bounding box **1010-3** and second bounding box **1010-4** corresponding to first object **1010-5** and second object **1010-6**, the first output image **1010-2** is sent to an crop unit **1000-2** configured to generate first cropped image **1010-7** and second cropped **1010-8**—associated with the first and second objects, respectively. The first and second cropped images are fed to embedding unit **1000-3** that generates a first object embedding **1010-11** and a second object embedding **1010-12**.

[0178] A matching unit **1000-4** identifies, for each one of the first object embedding and a second object embedding a matching reference embedding—first matching reference embedding **1012-1** and second matching reference embedding **1012-2**. If there is no match for any one of the object embeddings, then that object embedding is regarded to be being classified into an insufficient confidence level.

[0179] Assuming a first object related match—the first matching reference embedding **1012-1** represents a first reference embedding cluster **1013-1** associated with a first object classification **1014-1**—and the first object is classified as belonging to the first object classification **1014-1**.

[0180] Assuming a second object related match—the second matching reference embedding **1012-2** represents a second reference embedding cluster **1013-2** associated with a second object classification **1014-2**—and the second object is classified as belonging to the second object classification **1014-2**.

[0181] According to an embodiment, the matching and the clustering are not executed in the embedding domain but at the embedding signature domain. Accordingly—there is an embedding signature generator **1000-34** between the embedding unit **1000-3** and the matching unit **1004**. In this case the matching unit **1000-4** identifies, for each object embedding signature a matching reference embedding signature that represents a reference embedding signature cluster associated with an object classification **1014-1**.

[0182] It should be noted that the clusters are managed (re-evaluated, reduced, re-defined) by a cluster manager **1000-9**.

[0183] According to an embodiment, the object detection applied by the first neural network **1000-1** includes estimating a presence of an object within the sensed information unit. In addition to the bounding box, the first neural network also outputs its estimation of the object associated with the bounding box. This estimate is referred to as an initial object estimate. For example—first bounding box **1010-3** is associated with first object **1010-5** that is initially estimated to be a pedestrian.

[0184] According to an embodiment, system **101** generates first cropped image **1010-7** that includes the first object, generates a first object embedding **1010-11**, finds a first matching reference embedding **1012-1** that represents a first reference embedding cluster **1013-1** associated with a first object classification **1014-1**—classifies the first object as belonging to the first object classification **1014-1**.

[0185] According to an embodiment, when there is a mismatch between the first object classification **1014-1** and the initial estimate of the initial object estimate—system **101** determines that at least one of the initial object estimate or the first object embedding is faulty.

[0186] According to an embodiment, a mismatch occurs when the initial object estimate and the first object classification contradict each other—for example, the initial object estimate is of a pedestrian and the first object classification is of a vehicle.

[0187] According to an embodiment, the difference between the initial object estimate and the first object classification is not regarded as a mismatch. Such a difference may be attributed to the higher accuracy of the matching process—which may provide more details about the first object. For example—the initial object estimate may be coarser than the first object classification (may have coarser or broader classes than the finer classes used to determine the initial object estimate).

[0188] For example, the initial object estimate may provide an estimate of a type of an object (for example a vehicle) and the first object classification may determine a sub-type of the object (for example a minivan, a SUV, a station wagon, a vehicle manufactured by a certain manufacturer, a model of the vehicle).

[0189] According to an embodiment, when there is a mismatch, the system estimates that the initial object estimate is faulty—as the matching process is regarded to be more reliable than the initial object estimate.

[0190] According to an embodiment, the system is configured to respond to the mismatch.

[0191] According to an embodiment, the response includes at least one of: providing an indication of a first neural network error, requesting or suggesting to train the first neural network, requesting or suggesting or instructing to evaluate the embedding process, requesting or suggesting or instructing to evaluate the matching process, requesting or suggesting or instructing to evaluate the cropping process, requesting or suggesting or instructing to use a FP removal module (such as FP removal module **110** of FIG. **1**) to perform a FP removal operation, requesting or suggesting or instructing to perform a cluster management operation such as re-clustering or deleting a cluster or adding a cluster or changing one or more clustering parameter such as a distance between cluster members, a distance between members of different clusters, and the like.

[0192] System of one or more computers can be configured to perform particular operations or actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions. One general aspect includes a method for object classification. The method also includes receiving, by a processing circuit, a sensed information unit that includes information indicative of an object located within a vehicle environment. The method also includes dynamically generating, by the processing circuit, an embedding of the object, where the embedding is a discriminating feature vector representing the object. The method also includes comparing the embedding to a plurality of reference embedding clusters, where each of the plurality of reference embedding clusters is associated with an object classification of a plurality of reference objects. The method also includes classifying, based on the comparing step, the embedding as being associated with one of the plurality of reference embedding clusters. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

[0193] Implementations of the method for object classification may include one or more of the following features. The method may include limiting the dynamic range of the sensed information unit. The sensed information unit may include a cropped sensed information unit cropped from a sensed information unit received from a sensing unit of the vehicle. The supervised machine learning training may include applying a cost function that induces generation of similar reference embeddings to similar objects and dissimilar reference embeddings to dissimilar objects. The classifying step initiates further processing of the sensed information unit. The plurality of reference embedding clusters represents a larger group of reference embeddings, where the larger group of reference embeddings is generated during a supervised machine learning training may

include feeding cropped images to a bounding shape generating neural network. The method may include dynamically updating the plurality of reference embedding clusters. The plurality of reference embedding clusters is generated by clustering one or more subgroups of reference embeddings of a larger group of reference embeddings into a plurality of clusters.

[0194] A system of one or more computers can be configured to perform particular operations or actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions. One general aspect includes a method for travel lane feature classification. The method also includes obtaining, via a processing circuit, information indicative of a travel lane including one or more travel lane features located within a vehicle environment. The method also includes generating a plurality of keypoints from the information. The method also includes organizing the plurality of keypoints into one or more subgroups of keypoints, where each of the one or more subgroups of keypoints is indicative of one or more categories of travel lane features. The method also includes classifying, based on the organizing step, the one or more organized subgroups of keypoints as indicative of a travel lane marker. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

[0195] Implementations of the method for travel lane feature classification may include one or more of the following features. The obtaining step may include obtaining, by an imaging sensor, a field of view image of a viewable area including the environment of the vehicle. The organizing step may include clustering each of the one or more subgroups of keypoints according to one or more pre-determined aspect ratios. The method may include cropping the clustered one or more subgroups of keypoints by: determining minimum shape dimensions of a shape bounding the clustered one or more subgroups of keypoints; and adding a lateral margin and vertical margin to the minimum shape dimensions. The method may include rotating the cropped clustered one or more subgroups of keypoints to an upright position. The method may include resizing the upright cropped clustered one or more subgroups of keypoints to a fixed size. The organizing step may include clustering the one or more subgroups of keypoints based on aspect ratios. The organizing step may include training a classifier to classify each of the one or more subgroups of keypoints based on the aspect ratios. The method may include classifying, based on the organizing step, at least a second subgroup of keypoints as indicative of a road boundary. The method may include classifying, based on the organizing step, at least a third subgroup of keypoints as indicative of an incidental marking. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

[0196] FIG. 15 illustrates an example of a method **1100** for travel lane element classification.

[0197] According to an embodiment, method **1100** includes step **1110** of obtaining, via a processing circuit, information indicative of a travel lane including one or more travel lane elements located within the environment of a vehicle.

[0198] According to an embodiment, step **1110** is followed by step **1120** of generating a plurality of keypoints from the information.

[0199] According to an embodiment, step **1120** is followed by step **1130** of organizing the plurality of keypoints into one or more subgroups of keypoints, wherein each of the one or more subgroup of keypoints is indicative of one or more categories of travel lane elements.

[0200] According to an embodiment, step **1130** is followed by step **1140** of generating one or more embeddings of the one or more subgroup of keypoints.

[0201] According to an embodiment, step **1140** is followed by step **1150** of classifying, based on the one or more embeddings, the one or more organized subgroup of keypoints as indicative of a travel lane marker.

[0202] According to an embodiment, step **1150** is followed by step **1160** of responding to the classifying.

[0203] According to an embodiment, step **1160** may include at least one of: [0204] a. Triggering a determination of a driving related operation to be executed by the vehicle. The driving related operation may be a fully autonomous operation such as maintaining a current progress of a vehicle, changing a progress of the vehicle, whereas a progress refers to at least one of direction of propagation, speed of propagation and acceleration of the vehicle. Alternatively-the driving related operation is an ADAS operation such as suggesting a driving scheme operation of performing a more limited driving related operation (limited in the sense of its duration). [0205] b. Executing a driving related operation. [0206] c. Determining the navigation path of the vehicle. [0207] d. Populating a database with an outcome of the vehicle. [0208] e. Sending to one or more artificial intelligence agents information regarding the classification. [0209] f. Activating one or more artificial intelligence agents based on the classification. [0210] g. Further processing an initial (pre-cropping) sensed information unit. The cropped information unit was generated by cropping an initial (pre-cropping) information sensing unit.

[0211] According to an embodiment, method **1100** includes step **1145** of generating one or more signatures of the one or more embeddings, the one or more signatures are of higher dimensionality than the one or more embeddings, wherein the classifying is based on the one or more signatures.

[0212] Step **1150** of classifying the one or more organized subgroup of keypoints as indicative of a travel lane marker will be based on the signatures of the one or more embeddings.

[0213] According to an embodiment, the information indicative of the travel lane is a sensed information unit, and step **1140** may include or may be preceded by generating one or more cropped sensed information units, one cropped information unit per subgroup of keypoints. The one or more embeddings are generated based on the one or more cropped sensed information unit.

[0214] Any combination of any step of any method illustrated in the application is provided.

[0215] According to an embodiment there is provided a method that includes receiving, by a processing circuit, a sensed information unit that includes information indicative of an object located within a vehicle environment; dynamically generating, by the processing circuit, an embedding of the object, wherein the embedding is a discriminating feature vector representing the object; comparing the embedding to a plurality of reference embedding clusters, wherein each of the plurality of reference embedding clusters is associated with an object classification of a plurality of reference objects; and classifying, based on the comparing step, the embedding as being associated with one of the plurality of reference embedding clusters.

[0216] According to an embodiment the method further includes limiting the dynamic range of the sensed information unit.

[0217] According to an embodiment, the sensed information unit is a cropped sensed information unit.

[0218] According to an embodiment, the classifying step initiates further processing of the sensed information unit.

[0219] According to an embodiment, the plurality of reference embedding clusters represents a larger group of reference embeddings, wherein the larger group of reference embeddings is generated during a supervised machine learning training comprising feeding cropped images to a bounding shape generating neural network.

[0220] According to an embodiment, the supervised machine learning training comprises applying a cost function that induces generation of similar reference embeddings to similar objects and dissimilar reference embeddings to dissimilar objects.

[0221] According to an embodiment, the method includes dynamically updating the plurality of reference embedding clusters.

[0222] According to an embodiment, the plurality of reference embedding clusters is generated by clustering one or more subgroups of reference embeddings of a larger group of reference

embeddings into a plurality of clusters.

[0223] According to an embodiment there is provided a non-transitory computer readable medium for object classification, the non-transitory computer readable medium stores instructions that once executed by an object classification system of the vehicle cause the object classification system to: receive a sensed information unit that includes information indicative of an object located within a vehicle environment; dynamically generate an embedding of the object, wherein the embedding is a discriminating feature vector representing the object; compare the embedding to a plurality of reference embedding clusters, wherein each of the plurality of reference embedding clusters is associated with an object classification of a plurality of reference objects; and classify, based on the comparing step, the embedding as being associated with one of the plurality of reference embedding clusters.

[0224] According to an embodiment, the object classification system is further configured to limit the dynamic range of the sensed information unit.

[0225] According to an embodiment, the sensed information unit is a cropped sensed information unit.

[0226] According to an embodiment, the classifying step initiates further processing of the sensed information unit.

[0227] According to an embodiment, the plurality of reference embedding clusters represents a larger group of reference embeddings, wherein the larger group of reference embeddings is generated during a supervised machine learning training comprising feeding cropped images to a bounding shape generating neural network.

[0228] According to an embodiment, the supervised machine learning training comprises applying a cost function that induces generation of similar reference embeddings to similar objects and dissimilar reference embeddings to dissimilar objects.

[0229] According to an embodiment, the object classification system is further configured to dynamically update the plurality of reference embedding clusters.

[0230] According to an embodiment, the plurality of reference embedding clusters is generated by clustering one or more subgroups of reference embeddings of a larger group of reference embeddings into a plurality of clusters.

[0231] According to an embodiment there is provided an object classification system of a vehicle, the object classification system comprising: one or more processing circuits that comprise at least a part of an integrated circuit, the one or more processing circuits are configured to: receive a sensed information unit that includes information indicative of an object located within a vehicle environment; dynamically generate an embedding of the object, wherein the embedding is a discriminating feature vector representing the object; compare the embedding to a plurality of reference embedding clusters, wherein each of the plurality of reference embedding clusters is associated with an object classification of a plurality of reference objects; and classify, based on the comparing step, the embedding as being associated with one of the plurality of reference embedding clusters.

[0232] According to an embodiment, the plurality of reference embedding clusters represents a larger group of reference embeddings, wherein the larger group of reference embeddings is generated during a supervised machine learning training comprising feeding cropped images to a bounding shape generating neural network.

[0233] According to an embodiment, the supervised machine learning training comprises applying a cost function that induces generation of similar reference embeddings to similar objects and dissimilar reference embeddings to dissimilar objects.

[0234] According to an embodiment, the sensed information unit is a cropped sensed information unit.

[0235] In the foregoing detailed description, numerous specific details are set forth in order to provide a thorough understanding of the invention. However, it will be understood by those skilled

in the art that the present invention may be practiced without these specific details. In other instances, well-known methods, procedures, and components have not been described in detail so as not to obscure the present invention.

[0236] The subject matter regarding the invention is particularly pointed out and distinctly claimed in the concluding portion of the specification. The invention, however, both as to organization and method of operation, together with objects, features, and advantages thereof, may best be understood by reference to the following detailed description when read with the accompanying drawings.

[0237] It will be appreciated that for simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements.

[0238] Because the illustrated embodiments of the present invention may for the most part, be implemented using electronic components and circuits known to those skilled in the art, details will not be explained in any greater extent than that considered necessary as illustrated above, for the understanding and appreciation of the underlying concepts of the present invention and in order not to obfuscate or distract from the teachings of the present invention.

[0239] Any reference in the specification to a method should be applied mutatis mutandis to a device or system capable of executing the method and/or to a non-transitory computer readable medium that stores instructions for executing the method.

[0240] Any reference in the specification to a system or device should be applied mutatis mutandis to a method that may be executed by the system, and/or may be applied mutatis mutandis to non-transitory computer readable medium that stores instructions executable by the system.

[0241] Any reference in the specification to a non-transitory computer readable medium should be applied mutatis mutandis to a device or system capable of executing instructions stored in the non-transitory computer readable medium and/or may be applied mutatis mutandis to a method for executing the instructions.

[0242] Any combination of any module or unit listed in any of the figures, any part of the specification and/or any claims may be provided.

[0243] Any one of transformation module, active learning module, or clustering module, or any other module described herein, may be implemented in hardware and/or code, instructions and/or commands stored in a non-transitory computer readable medium, may be included in a vehicle, outside a vehicle, in a mobile device, in a server, and the like.

[0244] The vehicle may be any type of vehicle-such as a ground transportation vehicle, an airborne vehicle, or a water vessel.

[0245] The specification and/or drawings may refer to an image. An image is an example of a media unit. Any reference to an image may be applied mutatis mutandis to a media unit. A media unit may be an example of sensed information. Any reference to a media unit may be applied mutatis mutandis to any type of natural signal such as but not limited to signal generated by nature, signal representing human behavior, signal representing operations related to the stock market, a medical signal, financial series, geodetic signals, geophysical, chemical, molecular, textual and numerical signals, time series, and the like. Any reference to a media unit may be applied mutatis mutandis to sensed information. The sensed information may be of any kind and may be sensed by any type of sensors-such as a visual light camera, an audio sensor, a sensor that may sense infrared, radar imagery, ultrasound, electro-optics, radiography, LIDAR (light detection and ranging), etc. The sensing may include generating samples (for example, pixel, audio signals) that represent the signal that was transmitted, or otherwise reach the sensor.

[0246] The specification and/or drawings may refer to a spanning element. A spanning element may be implemented in software or hardware. Different spanning element of a certain iteration are

configured to apply different mathematical functions on the input they receive. Non-limiting examples of the mathematical functions include filtering, although other functions may be applied.

[0247] The specification and/or drawings may refer to a concept structure. A concept structure may include one or more clusters. Each cluster may include signatures and related metadata. Each reference to one or more clusters may be applicable to a reference to a concept structure.

[0248] The specification and/or drawings may refer to a processor. The processor may be a processing circuitry. The processing circuitry may be implemented as a central processing unit (CPU), and/or one or more other integrated circuits such as application-specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), full-custom integrated circuits, etc., or a combination of such integrated circuits.

[0249] Any combination of any steps of any method illustrated in the specification and/or drawings may be provided.

[0250] Any combination of any subject matter of any of claims may be provided.

[0251] Any combinations of systems, units, components, processors, sensors, illustrated in the specification and/or drawings may be provided.

[0252] Any reference to an object may be applicable to a pattern. Accordingly-any reference to object detection is applicable mutatis mutandis to a pattern detection.

[0253] A situation may be a singular location/combination of properties at a point in time. A scenario is a series of events that follow logically within a causal frame of reference. Any reference to a scenario should be applied mutatis mutandis to a situation.

[0254] The sensed information unit may be sensed by one or more sensors of one or more types. The one or more sensors may belong to the same device or system—or may belong to different devices of systems.

[0255] A perception unit may be provided and may be preceded by the one or more sensors and/or by one or more interfaces from receiving one or more sensed information units. The perception unit may be configured to receive a sensed information unit from an I/O interface and/or from a sensor. The perception unit may be followed by multiple narrow AI agents-also referred to as an ensemble of narrow AI agents.

[0256] A sensed information unit may or may not be processed before reaching the perception unit. Any processing may be providing-filtering, noise reduction, and the like.

Claims

1. A method that is computer implemented for downstream processing of embedding information items, the method comprises: receiving multiple evaluated element embedding information items that represent multiple evaluated elements within an environment of a vehicle; identifying that the multiple evaluated element embedding information items are classified into an insufficient confidence level; and for each one of the multiple evaluated embedding information items identified being classified into an insufficient confidence level, automatically routing evaluated element information to a corresponding embedding information item-based classification unit that is trained to classify elements represented by the evaluated element embedding information item associated with a corresponding population of embedding information items.
2. The method according to claim 1, wherein the identifying step comprises comparing the multiple element embedding information items to a plurality of reference embeddings information items that represent a plurality of reference embedding information items clusters.
3. The method according to claim 1, further comprising classifying the evaluated element, by the corresponding embedding information item-based classification unit, wherein the classifying triggers a determination of a driving related operation for a vehicle.
4. The method according to claim 1, wherein the evaluated element information is a sensed information unit.

5. The method according to claim 1, wherein the evaluated element information is a cropped sensed information unit.
6. The method according to claim 1, wherein the first embedding item-based classification unit is trained across a population of embedding that is larger than each corresponding population of embeddings.
7. The method according to claim 1, wherein the evaluated element embedding information item is an evaluated element embedding signature.
8. The method according to claim 1, wherein the evaluated element embedding information item is an evaluated element embedding.
9. The method according to claim 1, wherein the identifying triggers a generation of a routing rule for bypassing the step of comparing when receiving future evaluated element embedding information items having a same value as the evaluated element embedding information item.
10. The method according to claim 1, wherein the identifying triggers a re-evaluation of the plurality of reference embeddings information items.
11. The method according to claim 1, wherein the routing comprises selecting each corresponding embedding information item-based classification unit out of a plurality of embedding information item-based classification units.
12. A non-transitory computer readable medium for downstream processing of embedding information items, the non-transitory computer readable medium stores instructions that once executed by a computerized system cause the object computerized system to: receive multiple evaluated element embedding information items that represent multiple evaluated elements within an environment of a vehicle; identify that the multiple evaluated element embedding information items are classified into an insufficient confidence level; and for each one of the multiple evaluated embedding information items identified as being classified into an insufficient confidence level, automatically route evaluated element information to a corresponding embedding information item-based classification unit that is trained to classify elements represented by the evaluated element embedding information item associated with the corresponding population of embedding information items.
13. The non-transitory computer readable medium according to claim 12, wherein the identifying step comprises comparing the multiple element embedding information items to a plurality of reference embeddings information items that represent a plurality of reference embedding information items clusters.
14. The non-transitory computer readable medium according to claim 12, further storing instructions for classifying the evaluated element, by the corresponding embedding information item-based classification unit, wherein the classifying triggers a determination of a driving related operation for a vehicle.
15. The non-transitory computer readable medium according to claim 12, wherein the evaluated element information is a sensed information unit.
16. The non-transitory computer readable medium according to claim 12, wherein the evaluated element information is a cropped sensed information unit.
17. The non-transitory computer readable medium according to claim 12, wherein the first embedding item-based classification unit is trained across a population of embedding that is larger than each corresponding population of embeddings.
18. The non-transitory computer readable medium according to claim 12, wherein the identifying triggers a generation of a routing rule for bypassing the step of comparing when receiving future evaluated element embedding information items having a same value as the evaluated element embedding information item.
19. The non-transitory computer readable medium according to claim 12, wherein the identifying triggers a re-evaluation of the plurality of reference embeddings information items.
20. A computerized system for downstream processing of embedding information items, the

computerized system comprises: a memory unit that is configured to store multiple evaluated elements embedding information items that represents multiple evaluated elements within an environment of a vehicle; and a processing circuit that is configured to identify that the multiple evaluated element embedding information items are classified into an insufficient confidence level; and for each one of the multiple evaluated embedding information items identified as being classified into the insufficient confidence level, automatically routing the evaluated element embedding information item to a corresponding embedding information item-based classification unit that is trained to classify elements represented by the evaluated element embedding information item associated with the corresponding population of embedding information items.
